Antarctic Insights: A Statistical Analysis of the Palmer Penguins Dataset

Exploratory Data Analysis and Regression Modeling in Excel

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Executive Summary

This project involved a comprehensive analysis of the Palmer Penguins dataset using Microsoft Excel, leveraging tools such as descriptive statistical tables, correlation matrices, and boxplots to evaluate 342 biological records. Through initial data exploration and preparation, I identified that while Adelie penguins represent the largest population, Chinstrap penguins possess the longest average culmen length at 48.8 mm. To further understand these biological relationships, I developed a regression model that revealed a strong R^2 of 0.759, with an RMSE of 6.8939. This performance is highly indicative of a reliable model, as the standard error of 6.9133 represents only ~7% of the predicted data. The analysis highlighted a powerful positive correlation (0.8712) between body mass and flipper length, proving that body mass is a primary and significant predictor of physical dimensions within this Antarctic population.

DATASET

- This dataset contains data about the Palmer penguin's species in Antarctica. The dataset shows information related to which island, culmen size, flipper size, body mass, and sex of each type of penguin. The dataset contains seven main columns including: species, culmen length, culmen depth, flipper, body mass, island, and sex.

DATA EXPLORATION

Legend

Species Code: Adelie = 1
Chinstrap = 2
Gentoo = 3

Island Code: Torgersen = 1
Biscoe = 2
Dream = 3

Secody: Male = 1
Female = 0

	species	island	Culmen length mm	Culmen depth	Flipper length mm	Body mass g	sex
COUNT, cells with numbers	0	0	342	342	342	342	0
COUNTA, cells not empty	342	342	342	342	342	342	342
COUNTIF, cell with "NA"	0	0	0	0	0	0	0
AVERAGE, ignores "NA"	0	0	43.9	17.2	200.9	4201.8	0
STDEV, ignores "NA"	0	0	5.5	2.0	14.1	802.0	0
VAR	0	0	29.8	3.9	197.7	643131.1	0
MIN	0	0	32.1	13.1	172	2700	0
MAX	0	0	59.6	21.5	231	6300	0

Sample of the Data Set	Adelie	Torgersen	39.1	18.7	181	3750	MALE
Data type (Text, Integer or Decimal)	Text	Text	Decimal	Decimal	Integer	Integer	Text
Numerical or Categorical	Categorical	Categorical	Numerical	Numerical	Numerical	Numerical	Categorical
Continuous or Discrete	N/A	N/A	Continuous	Continuous	Discrete	Discrete	N/A
MODE, numerical fields only	1	2	41.1	17	190	3800	1

DATA PREPARATION

A	B	C	D	E	J	K	L	M	N	O	P	Q
studyName	Sample Number	Species	Region	Island	Culmen Length (mm)	Culmen Depth (mm)	Flipper Length (mm)	Body Mass (g)	Sex	Delta 15 N (o/oo)	Delta 13 C (o/oo)	Comments
PAL0708	1	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	39.1	18.7	181	3750	MALE			Not enough blood for isotopes.
PAL0708	2	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	39.5	17.4	186	3800	FEMALE	8.94956	-24.69454	
PAL0708	3	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	40.3	18	195	3250	FEMALE	8.36821	-25.33302	
PAL0708	4	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen								Adult not sampled.
PAL0708	5	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	36.7	19.3	193	3450	FEMALE	8.76651	-25.32426	
PAL0708	6	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	39.3	20.6	190	3650	MALE	8.66496	-25.29805	
PAL0708	7	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	38.9	17.8	181	3625	FEMALE	9.18718	-25.21799	Nest never observed with full clutch.
PAL0708	8	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	39.2	19.6	195	4675	MALE	9.4606	-24.89958	Nest never observed with full clutch.
PAL0708	9	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	34.1	18.1	193	3475				No blood sample obtained.
PAL0708	10	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	42	20.2	190	4250		9.13362	-25.09368	No blood sample obtained for sexing.
PAL0708	11	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	37.8	17.1	186	3300		8.63243	-25.21315	No blood sample obtained for sexing.
PAL0708	12	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	37.8	17.3	180	3700				No blood sample obtained.
PAL0708	13	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	41.1	17.6	182	3200	FEMALE			Not enough blood for isotopes.
PAL0708	14	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	38.6	21.2	191	3800	MALE			Not enough blood for isotopes.
PAL0708	15	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	34.6	21.1	198	4400	MALE	8.55583	-25.22588	
PAL0708	16	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	36.6	17.8	185	3700	FEMALE			Not enough blood for isotopes.
PAL0708	17	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	38.7	19	195	3450	FEMALE	9.18528	-25.06691	
PAL0708	18	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	42.5	20.7	197	4500	MALE	8.67538	-25.13993	
PAL0708	19	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	34.4	18.4	184	3325	FEMALE	8.47827	-25.23319	
PAL0708	20	Adelie Penguin (Pygoscelis adeliae)	Anvers	Torgersen	46	21.5	194	4200	MALE	9.11616	-24.77227	
PAL0708	21	Adelie Penguin (Pygoscelis adeliae)	Anvers	Biscoe	37.8	18.3	174	3400	FEMALE	8.73762	-25.09383	
PAL0708	22	Adelie Penguin (Pygoscelis adeliae)	Anvers	Biscoe	37.7	18.7	180	3600	MALE	8.66271	-25.0639	
PAL0708	23	Adelie Penguin (Pygoscelis adeliae)	Anvers	Biscoe	35.9	19.2	189	3800	FEMALE	9.22286	-25.03474	
PAL0708	24	Adelie Penguin (Pygoscelis adeliae)	Anvers	Biscoe	38.2	18.1	185	3950	MALE	8.43423	-25.22664	
PAL0708	25	Adelie Penguin (Pygoscelis adeliae)	Anvers	Biscoe	38.8	17.2	180	3800	MALE	9.63954	-25.29856	

Data Preparation and Remediation

To ensure the dataset was ready for accurate and meaningful analysis, a structured process was followed that included data preparation and remediation steps.

1. Attribute Selection

Relevant attributes were identified based on their potential contribution to addressing the research question. These include:

- Species
- Island
- Culmen Length
- Culmen Depth
- Flipper Length
- Body Mass
- Sex

2. Data Quality Checks

The dataset was examined for completeness and consistency:

- Counted non-empty cells and identified missing values (N/A).
- Verified data integrity by checking for duplicates and potential outliers.

3. Descriptive Statistics

For numerical attributes, key statistical measures were calculated to understand data distribution:

- Mean (excluding N/A values)
- Standard Deviation
- Variance
- Minimum and Maximum values
- Mode for numerical fields

These metrics provided insights into variability and central tendencies within the dataset.

4. Data Structure Review

A sample of the dataset was inspected to confirm:

- Data types (text, integer, decimal)
- Classification of variables as numerical or categorical
- Determination of whether numerical data was discrete or continuous

5. Encoding Categorical Variables

Categorical attributes such as species, island, and sex were assigned numeric codes to facilitate statistical analysis and modeling.

Data Issues and Remediation

During the preparation process, several issues were identified and addressed:

Missing Numerical Data

Some attributes, such as Body Mass, contained missing values. To resolve this:

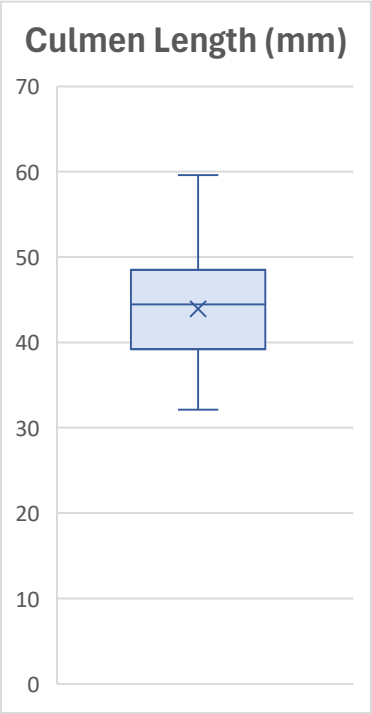
- Estimated missing values using the average for the same species.
- Created an additional column to store these estimated values for comparison with the original data, maintaining transparency in the imputation process.

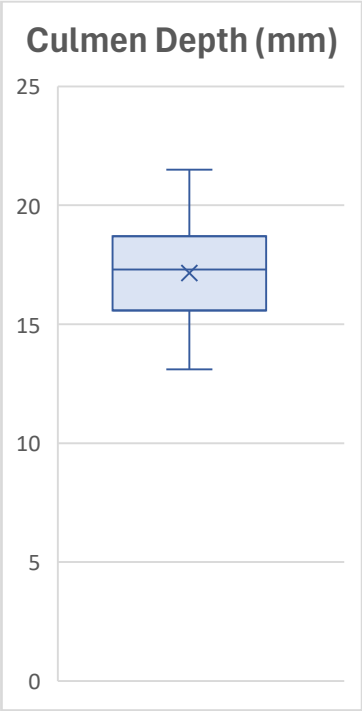
Rows with Multiple N/A Values

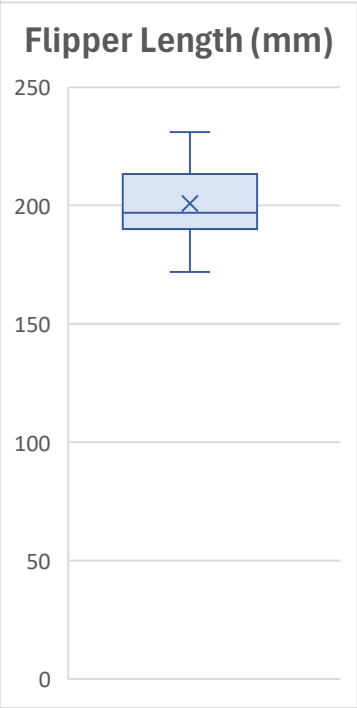
Certain observations contained multiple missing fields, making them irrelevant to the research questions and potentially skewing calculations.

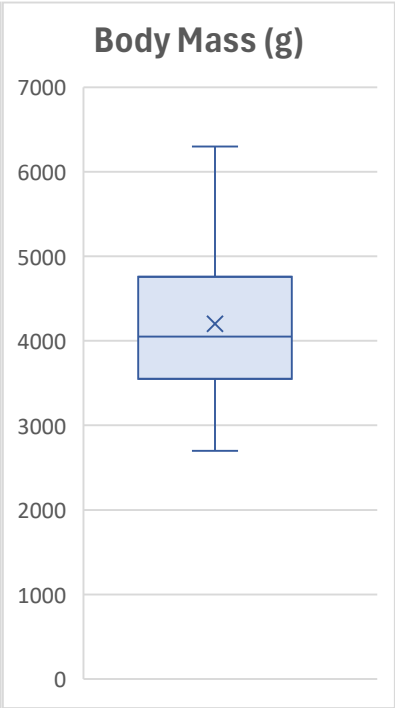
- These rows were removed from the dataset to preserve analytical accuracy and avoid introducing bias.

PRELIMINARY ANALYSIS









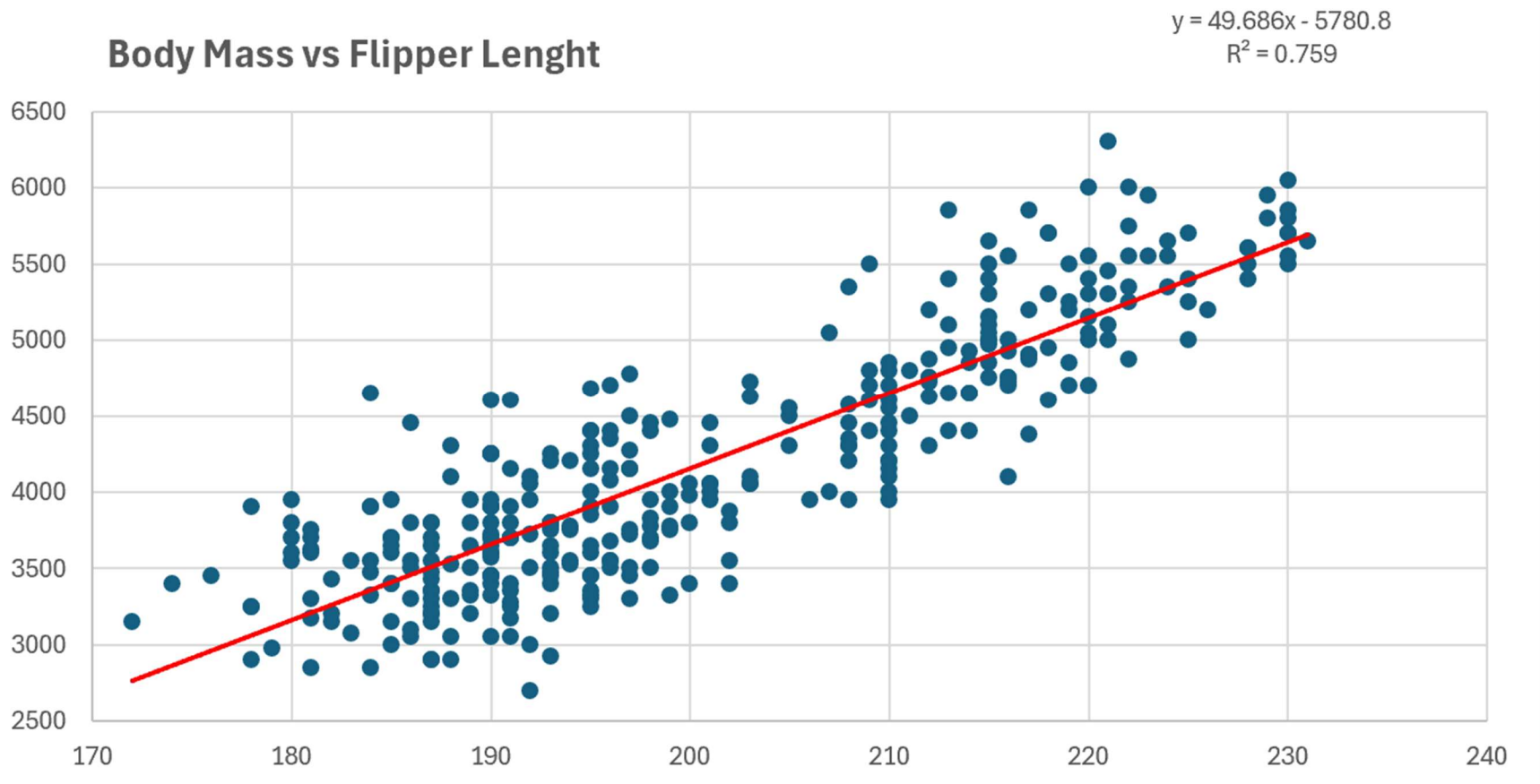
	culmen_length_mm	culmen_depth	flipper_length_mm	body_mass_g
Q1	39.225	15.6	190	3550
Q3	48.5	18.7	213	4750
IQR	9.275	3.1	23	1200
Effective Minimum	25.3125	10.95	155.5	1750
Effective Maximum	62.4125	23.35	247.5	6550

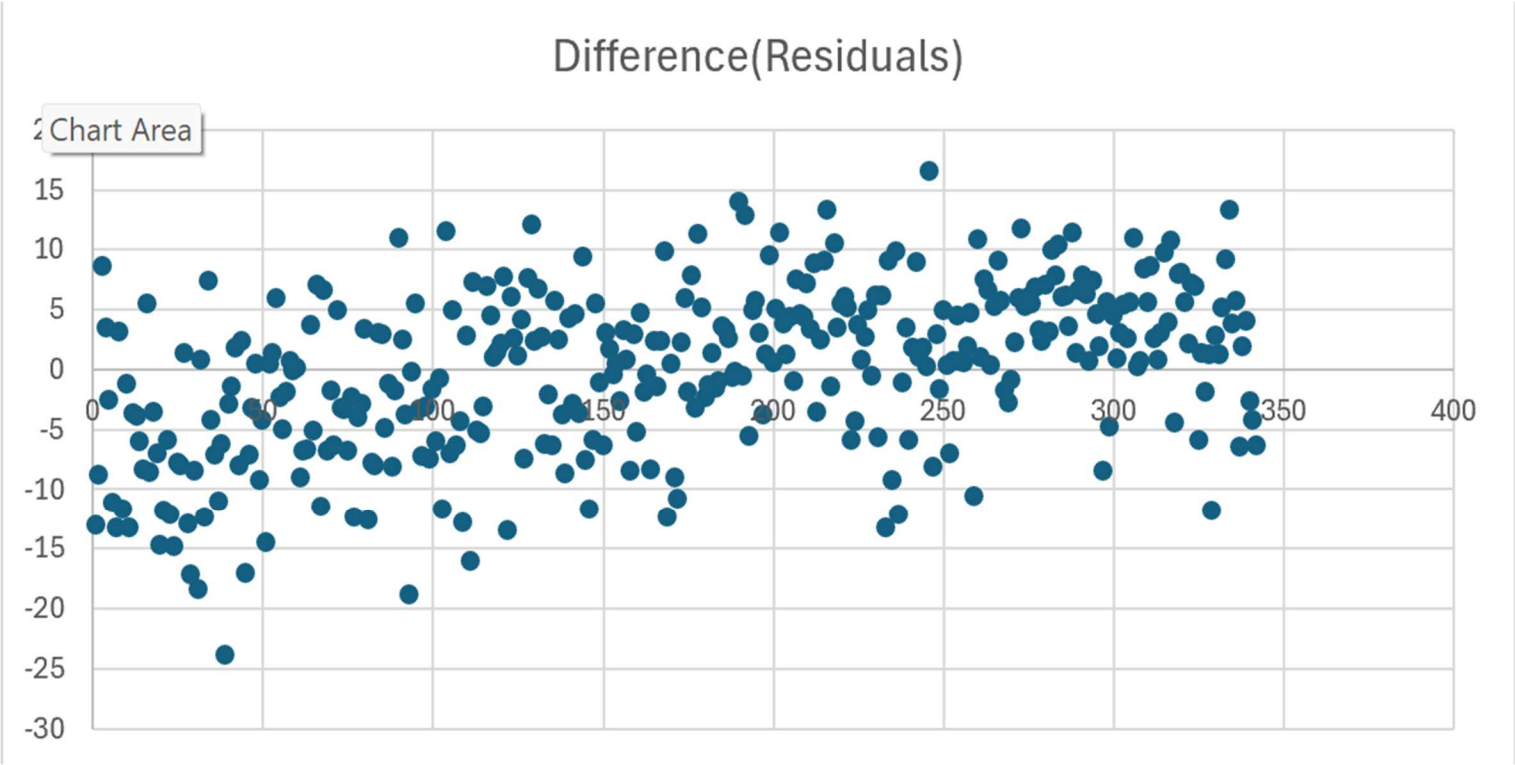
	species	island	Culmen length mm	Culmen depth mm	flipper length mm	Body mass g	sex
species	1	0.0047	0.731369143	-0.744076181	0.854307415	0.750491119	0.027859505
island	0.0047	1	0.223106091	0.179753381	-0.145432345	-0.189073796	-0.00120235
Culmen length mm	0.7314	0.2231	1	-0.23505287	0.656181341	0.595109824	0.347056479
Culmen depth mm	-0.744	0.1798	-0.23505287	1	-0.583851216	-0.471915621	0.34731926
Flipper length mm	0.8543	-0.145	0.656181341	-0.583851216	1	0.871201767	0.262741471
Body mass g	0.7505	-0.189	0.595109824	-0.471915621	0.871201767	1	0.429980683
sex	0.0279	-0.001	0.347056479	0.34731926	0.262741471	0.429980683	1

- The Culmen Length for all the three species of penguins is 43.9 mm on average.
- The Culmen depth for all the three species is 17,2 mm on average,

- The Flipper length for all three species is 200,9 mm on average,
- The body mass for the three species is 4,201,8 g (4.2 Kg) on average,
- The correlation table shows that Flipper Length and Body Mass have a positive and the strongest correlation. This is the pair I would like to explore more.

ANALYSIS


Body Mass vs Flipper Lenght
y = 49.686x - 5780.8
R² = 0.759
6500
6000
5500
5000
4500
4000
3500
3000
2500
170
180
190
200
210
220
230
240



- The R^2 confirms the strong correlation between the two variables (0.759), where the flipper length is strongly correlated to the body mass. This means that 76% of the variation of the flipper length can be explained by an increase of the body mass. The remaining 24% of the variation of the flipper length could be explained by the species type, since the correlation table shows that the pair species/flipper length has the second strongest correlation with 0.8543.

- The Standard Error of ~9% tells that predictive error is small relative to the data scale. A model with a standard error between 5-10% is considered good predictive model.

- The residuals do not show any specific pattern; they look randomly distributed.

CONCLUSIONS

1. The RMSE is 393.12 or ~9% deviation of the total, wish confirms that Flipper Length is an excellent predictor of body mass. A model with an RMSE within 5-10% is considered good.
2. The Mean of the Residuals is -0.11847. Since it is close to zero, it tells me that the model is not overpredicting or underpredicting.
3. The range of the flipper length is 92 mm, this is the difference between the Effective Min and the Effective Max.
4. An increase in the flipper length can largely be explained by an increase in body mass, which accounts for about 74% of the variation. The remaining 26% may be attributed to differences among species.

5. Culmen depth and species are the pair with the strongest negative correlation, -0.7440. Which can be due to species represented as numeric codes, where Adelie = 1, Chinstrap = 2, Gentoo = 3. Here Adelie has a deeper culmen while Gentoo has shallower culmen depth.
6. Adelie is the penguin species with bigger population, followed by Gentoo and then Chinstrap.
7. Chinstrap has the biggest Culmen on average with a size of 48.8 mm on average and Adelie has the smallest Culmen on Average, which reaches 38.8 mm on average. The average Culmen length for all the penguin species is 43.9mm.